# Meeting Notes: Exosuit-Assisted Learning, Contraction/Stability Ideas, and Lab Logistics

## Meeting Overview

These notes summarize a discussion about:

- modeling an exosuit/exoskeleton interaction with a human/manipulator,
- using stiffness/damping modulation as a feedback channel,
- contraction/stability-inspired control ideas,
- possible rehabilitation implications,
- simulation and implementation plans,
- paper direction and short-term milestones,
- and some separate lab/logistics discussion.

Because the source transcript was partially noisy, these notes organize the main ideas into a coherent technical summary while marking speculative or unclear parts accordingly.

## 1. Main Technical Theme

The central idea discussed was the following:

> Can an exosuit provide a physically meaningful feedback signal — especially through programmable stiffness and damping — that helps a weak or impaired controller (e.g., human motor system, simplified manipulator controller) learn more stable behavior more quickly?

The conversation framed this in several related ways:

- as a **control problem**,
- as a **learning problem**,
- as a **rehabilitation problem**,
- and as a **human–robot interaction problem**.

A recurring intuition was that if the exosuit shapes the interaction dynamics appropriately, then the user/controller may receive a *better feedback signal* than raw unstable motion alone.

## 2. Core Hypothesis

A working hypothesis from the meeting was:

> If an external device adds stabilizing structure (e.g., effective stiffness and damping), then a weak controller may learn faster because the interaction becomes more interpretable, more stable, and closer to a regime in which the controller can infer what corrective action is needed.

This was discussed in contrast to a situation where:

- the system is highly unstable,
- state changes are difficult to interpret,
- the mapping from observation to corrective action is unclear,
- and the user/brain receives only a vague "oscillation is bad" signal.

The exosuit was therefore viewed not only as an assistive actuator, but also as a **feedback-shaping interface**.

## 3. Discussion of Stiffness and Damping

A repeated point in the discussion was that the exosuit may provide:

- effective stiffness,
- effective damping,
- and possibly a more perceivable relation between state and force.

The group discussed whether the user may be able to *feel* changes corresponding to something like:

- higher stiffness ($K_P$-like effect),
- higher damping ($K_D$-like effect),
- or a more stable/critically damped response.

One interpretation was:

- oscillatory behavior might correspond to insufficient damping/stiffness,
- and the exosuit could provide a more directly interpretable signal,
- which may then help the user/controller adapt internal gains.

## 4. Relation to $K_P$ and $K_D$

The conversation frequently referred to proportional and derivative behavior, often informally as:

$$K_P, \quad K_D$$

and sometimes in the broader sense of "how stiff" and "how damped" the interaction should be.

A rough conceptual mapping discussed was:

- more stiffness $\leftrightarrow$ larger $K_P$,
- more damping $\leftrightarrow$ larger $K_D$.

However, it was also explicitly acknowledged that:

- the mapping is **not one-to-one**,
- physical exosuit parameters are not the same thing as controller gains,
- and learning from exosuit dynamics to internal controller parameters is a nontrivial transfer problem.

This was identified as one of the key conceptual challenges.

## 5. Contraction / Stability / CCM Discussion

There was discussion around contraction-based ideas and contraction control, including references to:

- contraction theory,
- CCM (control contraction metrics),
- residual control on top of an existing stabilizing controller,
- and the challenge that the input matrix $B$ may be state-dependent.

One concern raised was that in some contraction-control formulations, the input matrix $B$ is assumed to be state-independent, while in the current system the effective $B$ appears to depend on state.

That led to the observation that:

- a direct full-state contraction-based treatment may be difficult,
- local linearization around an equilibrium may be easier,
- and some current implementation resembles linearization or local residual control rather than a global contraction result.

A related idea was to think in terms of an equilibrium point and local behavior around it:

- define an equilibrium when the exosuit is inactive or at a nominal operating point,
- linearize there,
- and investigate whether contraction-like or stability-like properties can be imposed locally.

## 6. Simulation Discussion

There was substantial discussion about simulation behavior, including:

- plotting responses with exosuit on vs. off,
- adding disturbances,
- comparing sign-wave disturbances vs. step disturbances,
- and observing how quickly oscillations decay.

Several concrete simulation tasks were suggested:

1. Plot the system response with the exosuit active.
2. Plot the response with the exosuit inactive.
3. Overlay the plots on the same axes.
4. Use a simpler disturbance, possibly a step input, instead of a sinusoidal disturbance.
5. Evaluate how the response settles and whether damping improves.
6. Include any currently omitted mechanical effects (e.g., pendulum-like or passive terms) if they materially affect interpretation.

A recurring comment was that scale differences between plots can be misleading, so overlaying responses with consistent axes is important.

## 7. Mechanical/Modeling Interpretation

The exosuit interaction was discussed using a simplified mechanical analogy involving:

- springs,
- dampers,
- force perturbations,
- and coupled dynamics between the user/manipulator and the exosuit.

At a high level, the system was treated as something resembling a second-order dynamical system:

$$m\ddot{x} + c\dot{x} + kx = f$$

or, in a coupled interpretation, two such systems interacting through shared motion/force.

The broad idea was:

- the exosuit contributes effective mechanical parameters,
- the user/controller probes those parameters by interacting with the exosuit,
- and from the resulting state/force relationship, the system may infer how to adapt.

## 8. System Identification Idea

A major thread was an identification-style interpretation:

> Suppose the exosuit has some unknown effective stiffness and damping. By perturbing or pushing against it, observing motion and force, can the user/controller estimate those parameters and adapt accordingly?

This led to discussion of:

- probing the exosuit,
- observing state trajectories,
- measuring interaction forces,
- and regressing effective parameters such as stiffness and damping.

A simplified conceptual update law discussed informally was of the form:

$$K_{1,t+1} = K_{1,t} + \eta(\hat{K}_2 - K_{1,t})$$

where:

- $K_{1,t}$ is the current internal/controller-side stiffness estimate,
- $\hat{K}_2$ is an inferred exosuit/environment stiffness,
- and $\eta$ is an adaptation step size.

This was not presented as a final equation, but rather as a conceptual template for adaptation.

## 9. Interpretation as Motor Learning / Brain Adaptation

A key conceptual move in the discussion was to treat the manipulator/controller as an analog for the human motor system.

The proposed analogy was:

| **Simulation/Control Object** | **Human Interpretation** |
|---|---|
| Manipulator/controller | Human motor control / brain-driven control |
| Exosuit mechanical impedance | External assistive feedback shaping |
| Parameter estimation | Perceptual/motor learning from interaction |
| Gain adjustment | Motor adaptation / re-learning |
| Catch trials | Evidence of learned internal compensation |

The group discussed that humans do not literally learn by numbers like $K_P$ and $K_D$, but rather by:

- movement,
- force,

- perceived stiffness,
- perceived damping,
- and repeated interaction.

So the engineering model would estimate parameters numerically, but the biological interpretation is that the brain learns through repeated sensorimotor experience.

## 10. Rehabilitation Framing

The rehabilitation framing emerged strongly in the conversation.

The key idea was:

- after stroke or impairment, the necessary motor circuits may be weakened or partially lost,
- the body may still have some mechanical capability,
- but the controller (brain) no longer effectively produces the needed coordination,
- and an exosuit may provide both assistance and a structured training signal.

This led to the broader rehabilitation hypothesis:

> The exosuit may not only help complete the task in the moment, but may also help the nervous system re-learn appropriate control by providing stabilizing and informative interaction dynamics.

An important practical observation was that conventional rehabilitation is often:

- intermittent,
- clinic-based,
- and limited in training frequency.

So a wearable or home-usable system could potentially enable:

- more continuous exposure,
- more repetitions,
- and more persistent adaptation.

## 11. Difference from Passive Textile Devices

A question raised explicitly was:

> What is the difference between this approach and simply wearing a passive textile-based suit with some built-in stiffness?

The discussion suggested the following answer:

- A passive textile system provides a fixed or baseline mechanical effect.

- A cable/motor-driven exosuit can, in principle, provide **programmable** stiffness/damping.
- It may therefore shape different tasks differently.
- It can serve as an experimental platform for testing the theory before moving to lighter implementations.

At the same time, it was also acknowledged that:

- if a textile implementation can realize the same effective behavior,
- then from the scientific standpoint the implementation medium may be secondary.

In other words, the main contribution is the **hypothesis and mechanism**, not necessarily the first hardware embodiment.

## 12. Possible Experimental Structure

A possible staged validation plan emerged.

### 12.1. Stage 1: Simulation

- Build a simplified manipulator + exosuit model.
- Include effective stiffness and damping on the exosuit side.
- Probe the system with perturbations.
- Compare behavior with and without exosuit-induced stabilization.
- Test whether exosuit parameter estimation is feasible.
- Investigate whether estimated exosuit parameters can inform updates to weak controller gains.

### 12.2. Stage 2: Two-System Coupling Toy Model

A very simple conceptual model was proposed:

- system 1: weak learner/controller,
- system 2: better-behaved target/reference dynamics,
- couple them mechanically or through interaction,
- and ask whether system 1 can infer useful dynamic properties from the interaction.

This was discussed as a useful first-principles testbed.

### 12.3. Stage 3: Human Experiment

A rough human study concept was discussed:

- participants wear the exosuit,
- the exosuit imposes selected stiffness/damping settings,
- participants interact with or push against it,
- the brain experiences the altered dynamics,
- and catch trials are used to test whether participants learned an internal compensation.

A catch-trial logic was mentioned:

- if the exosuit suddenly removes the imposed effect,
- and the participant still produces a compensatory action,
- then that suggests the nervous system learned something about the altered dynamics.

## 13. Rehabilitation-Inspired Interpretation

We interpret the problem not merely as parameter estimation, but as a motor re-learning and compensation process inspired by post-stroke rehabilitation.

### 13.1. Conceptual Interpretation

After stroke, the brain's effective ability to regulate limb impedance may be impaired. In particular, the capacity to generate or modulate stiffness and damping can be reduced or poorly coordinated. An exosuit then provides external mechanical assistance. Through repeated interaction with this assistance, the nervous system may gradually infer the stabilizing mechanical support provided by the exosuit, and this experience can be used to reconstruct or compensate for lost internal impedance regulation.

In our robotic analogy:

- the manipulator corresponds to the arm,
- the CT controller corresponds to the brain,
- the CT stiffness and damping parameters correspond to internal neuromechanical regulation,
- the exosuit corresponds to an external assistive device,
- probing interaction corresponds to exploratory motor behavior,
- parameter updates in the CT controller correspond to motor re-learning or adaptation.

Thus, the central hypothesis is that an impaired controller can recover useful impedance regulation by interacting with an external assistive impedance and internalizing its mechanical characteristics.

### 13.2. Mechanical Model

**State and equations of motion.** The implemented system is a 2-DOF cable-driven manipulator (`manipulator_cable_exo_springs_elbow_follow` URDF) with a Series Elastic Actuator (SEA) on the elbow (joint 2) and rigid direct-drive at the shoulder (joint 1). The full system state is

$$x = [q_1,\, q_2,\, \dot{q}_1,\, \dot{q}_2,\, \theta_m,\, \dot{\theta}_m]^\top \in \mathbb{R}^6, \tag{1}$$

where $q_1$ is the shoulder angle, $q_2$ is the elbow angle, and $(\theta_m, \dot{\theta}_m)$ are the drive-motor position and velocity at the joint-2 SEA cable motor. The 2R manipulator equations of motion are

$$M(q_2)\,\ddot{q} = -C(q,\dot{q})\,\dot{q} - G(q) + \tau, \tag{2}$$

with $q = [q_1, q_2]^\top$ ($m_1 = 5.31\,\mathrm{kg}$, $L_1 = 0.335\,\mathrm{m}$; $m_2 = 0.52\,\mathrm{kg}$, $L_2 = 0.19\,\mathrm{m}$) and applied joint torques

$$\tau_1 = \tau_{\mathrm{CT},1}, \tag{3}$$

$$\tau_2 = r_p\, F_{\mathrm{cable}} + \tau_{\mathrm{exo}} - b_{j2}\, \dot{q}_2. \tag{4}$$

**SEA cable dynamics.** The joint-2 cable routes from the motor drum through a spring to the big pulley ($r_p = 0.04775\,\mathrm{m}$, gear ratio $N = 6$). The spring extension and cable force are

$$\delta = r_p\left(\frac{\theta_m}{N} - q_2\right), \qquad F_{\text{cable}} = k_s\,\delta + b_c\,\dot{\delta}, \tag{5}$$

and the motor obeys 2nd-order rotor dynamics

$$J_m\,\ddot{\theta}_m = \frac{\tau_{2,\text{des}}}{N} - b_m\,\dot{\theta}_m - \frac{r_p\,F_{\text{cable}}}{N}. \tag{6}$$

Default parameters: $k_s = 300\,\mathrm{N/m}$, $b_c = 2.0\,\mathrm{N{\cdot}s/m}$, $J_m = 3.32 \times 10^{-5}\,\mathrm{kg{\cdot}m^2}$.

**Exosuit model (Method B, centred elbow pulley).** Two antagonistic exo cables wrap a pulley centred on the elbow axis. In *deactivated* (transparent) mode the exo motors track the elbow encoder so spring extensions remain zero. In *activated* (co-contraction) mode a symmetric angular offset $\Delta\theta$ is added to both motor commands:

$$\frac{\theta_{mR}}{N} = q_2 + \Delta\theta, \qquad \frac{\theta_{mL}}{N} = -q_2 + \Delta\theta. \tag{7}$$

The resulting spring extensions are

$$\delta_R = r_{\text{exo}}\left(\frac{\theta_{mR}}{N} - q_2\right), \qquad \delta_L = r_{\text{exo}}\left(\frac{\theta_{mL}}{N} + q_2\right), \tag{8}$$

and the net exo torque (sign convention: virtual work on $q_2$) is

$$\tau_{\text{exo}} = r_{\text{exo}}\,(F_R - F_L), \qquad F_i = k_{\text{exo}}\,\delta_i + b_{\text{exo}}\,\dot{\delta}_i. \tag{9}$$

Linearised about the activation anchor $q_{2,a}$ this reduces to

$$\tau_{\text{exo}} \approx -K_{\text{exo,eff}}\,(q_2 - q_{2,a}) - D_{\text{exo,eff}}\,\dot{q}_2, \tag{10}$$

with effective joint-level stiffness and damping

$$K_{\text{exo,eff}} = 2\,k_{\text{exo}}\,r_{\text{exo}}^2, \qquad D_{\text{exo,eff}} = 2\,b_{\text{exo}}\,r_{\text{exo}}^2. \tag{11}$$

For the reference parameters ($r_{\text{exo}} = 0.04775\,\mathrm{m}$, $k_{\text{exo}} = 8000\,\mathrm{N/m}$, $b_{\text{exo}} = 2.0\,\mathrm{N{\cdot}s/m}$):

$$K_{\text{exo,eff}} = 2 \times 8000 \times 0.04775^2 \approx 36.5\ \mathrm{Nm/rad}, \qquad D_{\text{exo,eff}} \approx 0.009\ \mathrm{Nm\,s/rad}. \tag{12}$$

**Impairment and assisted behavior.** Let the impaired CT controller initially have reduced elbow impedance:

$$K_c^{\text{imp}} < K_c^{\text{healthy}}, \qquad B_c^{\text{imp}} < B_c^{\text{healthy}}. \tag{13}$$

When the exosuit is active, the total effective elbow impedance becomes

$$K_{\text{eff}} = K_c^{\text{imp}} + K_{\text{exo,eff}}, \qquad B_{\text{eff}} = B_c^{\text{imp}} + D_{\text{exo,eff}}, \tag{14}$$

so the exosuit externally compensates for the lost internal impedance while the motor-state and cable dynamics remain active.

### 13.3. Learning and Internalization

The controller probes the exosuit through repeated interaction and estimates its impedance parameters:

$$\hat{K}_e, \qquad \hat{B}_e. \tag{15}$$

These learned quantities are then used to update the weakened internal impedance parameters of the CT controller:

$$K_c^{\text{new}} = K_c^{\text{imp}} + \alpha_K \hat{K}_e, \tag{16}$$

$$B_c^{\text{new}} = B_c^{\text{imp}} + \alpha_B \hat{B}_e, \tag{17}$$

where $\alpha_K$ and $\alpha_B$ are internalization gains.

These equations express the key idea of the framework: the controller does not merely compensate online for the exosuit, but instead absorbs some of the exosuit's effective mechanical support into its own internal control structure.

Two interpretations are possible.

**Interpretation A: Direct compensation.** The controller adds back some of the lost stiffness and damping directly according to

$$K_c \leftarrow K_c + \alpha_K \hat{K}_e, \qquad B_c \leftarrow B_c + \alpha_B \hat{B}_e. \tag{18}$$

**Interpretation B: Internal model learning.** The controller uses the exosuit as a template for stabilizing impedance and gradually reconstructs an internal control law inspired by the support it experiences. In this interpretation, the exosuit acts not only as an assistive device but also as a teaching signal.

### 13.4. Role of Probing

In this framework, probing is not simply a technical procedure for identifying a passive environment. Instead, probing represents exploratory interaction through which the impaired controller acquires missing mechanical knowledge. Repeated probing allows the system to infer the exosuit's effective stiffness and damping and then use these inferred properties to restore its own reduced impedance regulation.

A suitable summary statement is:

> The manipulator performs repeated exploratory interactions with the exosuit, analogous to how a stroke-impaired arm may use assisted movement experience to infer stabilizing mechanical support. These interactions reveal the exosuit's effective stiffness and damping, which are then used to update the weakened internal impedance parameters of the CT controller.

### 13.5. Earlier Approaches Revisited Under This Framing

**1. Direct identification.** The manipulator estimates exosuit stiffness and damping from force-motion data. Under the rehabilitation-inspired interpretation, this corresponds to sensing the mechanical support provided by the exosuit. However, identification alone does not explain how the controller re-learns.

**2. Recursive or adaptive estimation.** The manipulator continuously updates $\hat{K}_e$ and $\hat{B}_e$ during interaction. This corresponds naturally to gradual motor learning through repeated assisted movement and fits the rehabilitation narrative better than one-shot identification.

**3. Active probing or experiment design.** The manipulator chooses informative motions that reveal exosuit impedance. For example, slow movements may reveal stiffness, while oscillatory motions may reveal damping. This corresponds to exploratory movement patterns that help the impaired system sense assistance more effectively.

**4. Adaptive controller update.** After estimating exosuit parameters, the CT controller updates its internal stiffness and damping. This is the central rehabilitation mechanism in the framework:

$$K_{CT}(t+1) = K_{CT}(t) + \eta_K\big(\hat{K}_e(t) - K_{CT}(t)\big), \tag{19}$$

$$B_{CT}(t+1) = B_{CT}(t) + \eta_B\big(\hat{B}_e(t) - B_{CT}(t)\big), \tag{20}$$

or, in additive form,

$$K_{CT}(t+1) = K_{CT}(t) + \eta_K \hat{K}_e(t), \tag{21}$$

$$B_{CT}(t+1) = B_{CT}(t) + \eta_B \hat{B}_e(t). \tag{22}$$

**5. Uncertainty-aware learning.** The controller may maintain confidence in its impedance estimates and update itself only when the estimates are reliable:

$$K_{CT}(t+1) = K_{CT}(t) + \eta_K\, w_t\, \hat{K}_e(t), \tag{23}$$

$$B_{CT}(t+1) = B_{CT}(t) + \eta_B\, w_t\, \hat{B}_e(t), \tag{24}$$

where $w_t$ is a confidence-dependent weighting factor.

### 13.6. Reinforcement Learning Under This Interpretation

Under this rehabilitation-inspired framing, reinforcement learning can be interpreted in several ways.

**RL for probing policy.** The agent learns which exploratory motions best reveal exosuit stiffness and damping. This corresponds to learning how to probe assistance efficiently and safely. A representative reward is

$$r_t = \alpha\big(\mathrm{tr}(P_t) - \mathrm{tr}(P_{t+1})\big) - \beta\|\tau_t\|^2 - \gamma\,\mathrm{safetyCost}_t, \tag{25}$$

where $P_t$ is an uncertainty covariance matrix.

**RL for CT parameter update.** Instead of prescribing the update law analytically, reinforcement learning chooses how much stiffness and damping to internalize based on movement quality, interaction effort, and stability.

**RL for probing and internalization jointly.** A more complete formulation allows the agent both to probe the exosuit and to update internal controller parameters:

$$s_t = [q, \dot{q}, \hat{K}_e, \hat{B}_e, K_{CT}, B_{CT}, \text{uncertainty}, e], \tag{26}$$

$$a_t = [\text{probe command}, \Delta K_{CT}, \Delta B_{CT}], \tag{27}$$

with reward

$$r_t = -\alpha e_t^2 - \beta \|\tau_t\|^2 - \gamma\, \text{safetyCost} + \delta\, \text{infoGain} + \zeta\, \text{internalRecovery}. \tag{28}$$

### 13.7. Best Practical Methodology

A practical methodology consistent with this framework is:

1. Initialize the CT controller with weakened impedance:

$$K_{CT} = K_{\text{weak}}, \qquad B_{CT} = B_{\text{weak}}. \tag{29}$$

2. Attach an exosuit with unknown impedance parameters $K_e$ and $B_e$.
3. Let the manipulator probe the exosuit repeatedly and estimate $\hat{K}_e$ and $\hat{B}_e$.
4. Update the CT controller according to

$$K_{CT} \leftarrow K_{CT} + \alpha_K \hat{K}_e, \qquad B_{CT} \leftarrow B_{CT} + \alpha_B \hat{B}_e. \tag{30}$$

5. Gradually reduce exosuit contribution and test whether the CT controller retains improved behavior after exosuit removal.

### 13.8. Experimental Narrative

This framework suggests the following sequence of experiments:

1. impaired controller only,
2. impaired controller with exosuit assistance,
3. impaired controller with exosuit assistance and learning,
4. exosuit removal after learning.

The key question is whether the controller retains improved stiffness and damping regulation after the exosuit is removed, analogous to retention of motor improvements after rehabilitation training.

### 13.9. Summary

We therefore treat the exosuit not only as a support device, but as a teaching mechanism. By repeatedly probing and estimating the exosuit's effective stiffness and damping, the manipulator updates the weakened stiffness and damping parameters of the CT controller. This provides a robotic analogue of post-stroke motor re-learning, in which external mechanical assistance is gradually internalized into improved intrinsic control.

## 14. Formal Problem Statement

We consider a manipulator interacting with an assistive exosuit at a single joint, taken here to be the elbow for concreteness. The manipulator is controlled by a CT controller whose intrinsic impedance parameters are intentionally weakened to model post-stroke impairment. The exosuit provides external stiffness and damping, and the manipulator is allowed to probe this assistance through repeated interaction.

The objective is not only to estimate the exosuit impedance, but to use the estimated exosuit mechanics to restore the weakened internal impedance parameters of the CT controller.

### 14.1. Plant and Interaction Model

The actual implemented system has a 6D state (see Eq. (1))

$$x = [q_1,\, q_2,\, \dot{q}_1,\, \dot{q}_2,\, \theta_m,\, \dot{\theta}_m]^\top \in \mathbb{R}^6, \tag{31}$$

where $q_1$ is the shoulder angle, $q_2$ the elbow angle, and $(\theta_m, \dot{\theta}_m)$ the SEA drive-motor states. For the learning and identification formulation below we focus on the elbow interaction coordinate $q \triangleq q_2$.

The exosuit torque at the elbow (linearised form, Eq. (10)) is

$$\tau_{\text{exo}}(t) = -K_e\big(q_2(t) - q_{2,a}\big) - B_e\,\dot{q}_2(t), \tag{32}$$

where the parameters to be identified are tied to physical cable quantities:

$$K_e \equiv K_{\text{exo,eff}} = 2\,k_{\text{exo}}\,r_{\text{exo}}^2, \qquad B_e \equiv D_{\text{exo,eff}} = 2\,b_{\text{exo}}\,r_{\text{exo}}^2, \tag{33}$$

and $q_{2,a}$ is the elbow angle captured at the moment of exo activation.

The CT controller has internal impedance parameters $K_{CT}(t)$ and $B_{CT}(t)$, initialized at impaired values

$$K_{CT}(0) = K_{\text{weak}}, \qquad B_{CT}(0) = B_{\text{weak}}. \tag{34}$$

The elbow joint effective torque with exosuit assistance is (abstracting the SEA-cable and motor dynamics to an equivalent direct-drive for the impedance view)

$$\tau_{\text{tot}}(t) \approx -\big(K_{CT}(t) + K_e\big)\big(q_2(t) - q_{2,a}\big) - \big(B_{CT}(t) + B_e\big)\dot{q}_2(t), \tag{35}$$

confirming that exosuit assistance temporarily fills the impedance deficit. The full 6D dynamics including SEA cable lag are captured by Eqs. (2)–(6).

### 14.2. Estimation Objective

During repeated probing, the manipulator observes force-motion data and estimates the exosuit impedance:

$$\hat{\theta}(t) = \begin{bmatrix} \hat{K}_e(t) \\ \hat{B}_e(t) \end{bmatrix}, \qquad \theta = \begin{bmatrix} K_e \\ B_e \end{bmatrix}. \tag{36}$$

The identification problem is to infer $\theta$ from interaction data of the form

$$y_t = \phi_t^\top \theta + \varepsilon_t, \tag{37}$$

where

$$y_t = \tau_{\text{exo}}(t), \qquad \phi_t = \begin{bmatrix} -(q(t) - q_0) \\ -\dot{q}(t) \end{bmatrix}, \tag{38}$$

and $\varepsilon_t$ represents modeling and measurement error.

### 14.3. Internalization Objective

The rehabilitation-inspired objective is to use the learned exosuit impedance to update the weakened CT controller:

$$K_{CT}(t+1) = \mathcal{F}_K\big(K_{CT}(t), \hat{K}_e(t)\big), \qquad B_{CT}(t+1) = \mathcal{F}_B\big(B_{CT}(t), \hat{B}_e(t)\big), \tag{39}$$

so that the controller gradually reconstructs lost intrinsic impedance regulation.

### 14.4. Overall Objective

The overall objective is to design:

1. a probing policy that generates informative and safe interactions,
2. an estimator that recovers $\hat{K}_e$ and $\hat{B}_e$,
3. an update law that transfers learned exosuit impedance into CT controller parameters,

such that:

- movement quality improves during assistance,
- the CT controller internalizes useful stiffness and damping,
- improved behavior is retained, at least partially, after exosuit removal.

### 14.5. Retention Criterion

A central test of the framework is performance after exosuit removal. Let the exosuit be removed after a training phase, so that $\tau_{\text{exo}} = 0$. The controller is then evaluated using its updated internal parameters $K_{CT}^{\text{post}}$ and $B_{CT}^{\text{post}}$. The method is successful if

$$J_{\text{post}} < J_{\text{weak}}, \tag{40}$$

where $J_{\text{post}}$ is a post-training performance cost and $J_{\text{weak}}$ is the cost of the initial weakened controller. This establishes that learning has been internalized rather than merely borrowed from the exosuit online.

## 15. Derivation of CT Parameter Update Laws

### 15.1. Motivation

The weakened CT controller is interpreted as having lost part of its intrinsic impedance regulation. The exosuit provides external stiffness and damping that compensate for this deficit during interaction. By estimating the exosuit impedance and transferring some fraction of it into the controller, we seek to reconstruct lost internal impedance.

### 15.2. Impaired and Assisted Impedance

Let the desired healthy internal impedance be

$$K_{\text{healthy}}, \qquad B_{\text{healthy}}. \tag{41}$$

Let the impaired controller start with

$$K_{CT}(0) = K_{\text{weak}}, \qquad B_{CT}(0) = B_{\text{weak}}, \tag{42}$$

where

$$K_{\text{weak}} < K_{\text{healthy}}, \qquad B_{\text{weak}} < B_{\text{healthy}}. \tag{43}$$

When the exosuit is attached, the total effective impedance becomes approximately

$$K_{\text{eff}}(t) = K_{CT}(t) + K_e, \qquad B_{\text{eff}}(t) = B_{CT}(t) + B_e. \tag{44}$$

Since the controller does not know $K_e$ and $B_e$ a priori, it forms estimates $\hat{K}_e(t)$ and $\hat{B}_e(t)$ through interaction.

### 15.3. Additive Internalization Rule

The simplest update law assumes that the controller internalizes a fraction of the estimated exosuit impedance:

$$K_{CT}(t+1) = K_{CT}(t) + \alpha_K \hat{K}_e(t), \tag{45}$$

$$B_{CT}(t+1) = B_{CT}(t) + \alpha_B \hat{B}_e(t), \tag{46}$$

where $0 < \alpha_K \leq 1$ and $0 < \alpha_B \leq 1$ are learning gains.

This law is appropriate when the exosuit is viewed as providing a mechanically meaningful template whose contribution can be added back into internal control.

### 15.4. Error-Driven Internalization Rule

A more conservative rule updates the controller toward the estimated exosuit impedance:

$$K_{CT}(t+1) = K_{CT}(t) + \eta_K \big(\hat{K}_e(t) - K_{CT}(t)\big), \tag{47}$$

$$B_{CT}(t+1) = B_{CT}(t) + \eta_B \big(\hat{B}_e(t) - B_{CT}(t)\big), \tag{48}$$

with $0 < \eta_K < 1$ and $0 < \eta_B < 1$.

This can be interpreted as a first-order adaptation law. Rearranging,

$$K_{CT}(t+1) = (1-\eta_K) K_{CT}(t) + \eta_K \hat{K}_e(t), \tag{49}$$

$$B_{CT}(t+1) = (1-\eta_B) B_{CT}(t) + \eta_B \hat{B}_e(t). \tag{50}$$

Thus the controller moves gradually toward a learned impedance target.

### 15.5. Deficit-Based Recovery Rule

If a desired healthy target impedance is available, define the impairment deficit as

$$\Delta K_{\mathrm{def}}(t) = K_{\mathrm{healthy}} - K_{CT}(t), \qquad \Delta B_{\mathrm{def}}(t) = B_{\mathrm{healthy}} - B_{CT}(t). \tag{51}$$

Then one may internalize only the amount needed to close the deficit:

$$K_{CT}(t+1) = K_{CT}(t) + \alpha_K \min\big(\hat{K}_e(t), \Delta K_{\mathrm{def}}(t)\big), \tag{52}$$

$$B_{CT}(t+1) = B_{CT}(t) + \alpha_B \min\big(\hat{B}_e(t), \Delta B_{\mathrm{def}}(t)\big). \tag{53}$$

This prevents over-compensation and is useful when healthy reference values are known or estimated.

### 15.6. Confidence-Weighted Update

Because the exosuit estimates may be uncertain, the update can be weighted by confidence. Let $w_t \in [0, 1]$ denote confidence, for example derived from an estimator covariance $P_t$. Then

$$K_{CT}(t+1) = K_{CT}(t) + \alpha_K w_t \hat{K}_e(t), \tag{54}$$

$$B_{CT}(t+1) = B_{CT}(t) + \alpha_B w_t \hat{B}_e(t). \tag{55}$$

A typical choice is

$$w_t = \exp\big(-\lambda \, \mathrm{tr}(P_t)\big), \tag{56}$$

or any monotone function that increases as uncertainty decreases.

### 15.7. Retention-Oriented Update

To model persistence after exosuit removal, one may include a retention state. Let $K_{\mathrm{mem}}(t)$ and $B_{\mathrm{mem}}(t)$ denote learned internal memory terms:

$$K_{\mathrm{mem}}(t+1) = \rho_K K_{\mathrm{mem}}(t) + \alpha_K w_t \hat{K}_e(t), \tag{57}$$

$$B_{\mathrm{mem}}(t+1) = \rho_B B_{\mathrm{mem}}(t) + \alpha_B w_t \hat{B}_e(t), \tag{58}$$

where $0 \leq \rho_K, \rho_B \leq 1$ are retention factors. Then

$$K_{CT}(t) = K_{\mathrm{weak}} + K_{\mathrm{mem}}(t), \qquad B_{CT}(t) = B_{\mathrm{weak}} + B_{\mathrm{mem}}(t). \tag{59}$$

This makes explicit the idea that the controller stores learned impedance beyond immediate interaction.

### 15.8. Recommended Update Law

For a first implementation, a good compromise is the confidence-weighted additive law:

$$K_{CT}(t+1) = K_{CT}(t) + \alpha_K w_t \hat{K}_e(t), \tag{60}$$

$$B_{CT}(t+1) = B_{CT}(t) + \alpha_B w_t \hat{B}_e(t), \tag{61}$$

with saturation constraints

$$K_{CT}(t+1) \in [K_{\min}, K_{\max}], \qquad B_{CT}(t+1) \in [B_{\min}, B_{\max}]. \tag{62}$$

This update law is simple, interpretable, biologically plausible, and directly aligned with the idea of internalizing reliable external support.

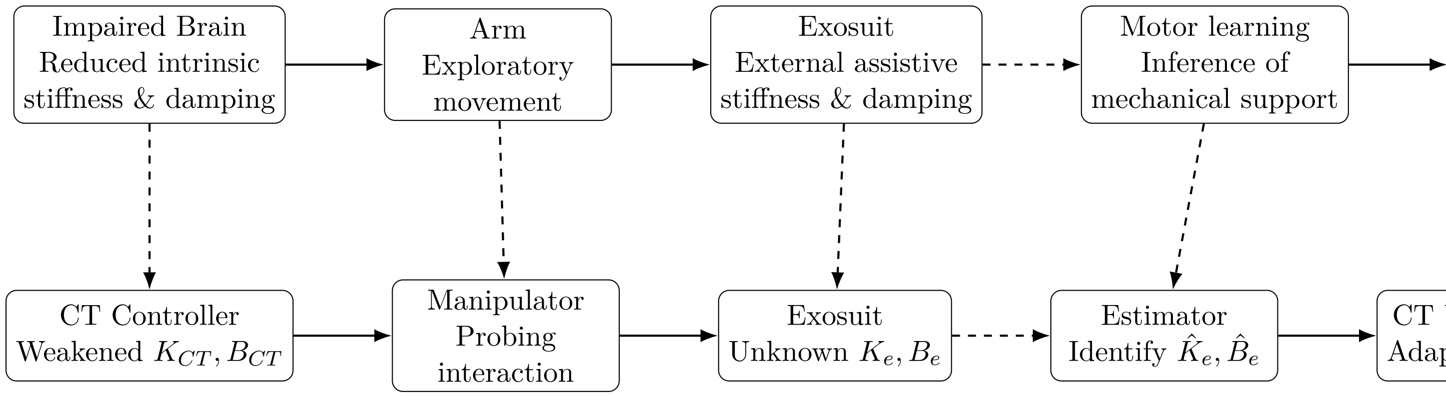


Figure 1: Analogy between biological post-stroke rehabilitation and the proposed robotic learning framework. The impaired brain corresponds to a CT controller with weakened impedance parameters. The exosuit acts both as assistance and as a teaching mechanism, whose estimated stiffness and damping are internalized into the controller.

## 16. Block-Diagram Analogy

## 17. Reinforcement Learning for Probing Strategy

### 17.1. Motivation

A key question in the proposed framework is how the manipulator should probe the exosuit in order to estimate its impedance efficiently, safely, and with minimal unnecessary effort. If the probing motion is too weak, the exosuit parameters may not be sufficiently excited for accurate estimation. If the probing motion is too aggressive, the interaction may become unsafe, energetically inefficient, or inconsistent with rehabilitation-inspired behavior.

We therefore formulate probing as a sequential decision-making problem and use reinforcement learning (RL) to learn an informative probing policy.

### 17.2. Role of RL in the Framework

In this framework, RL is not primarily used to replace the CT controller. Instead, RL determines the *probing strategy*, that is, the exploratory motion or excitation pattern that the manipulator should generate in order to reveal the exosuit's effective stiffness and damping.

Thus:

- the CT controller governs movement execution and impedance-based interaction,
- the estimator identifies exosuit parameters from measured data,
- the RL agent selects probing actions that improve parameter identifiability.

This interpretation is consistent with the rehabilitation analogy: the system learns *how to explore assistance* in a way that reveals useful mechanical information.

### 17.3. RL Formulation

We model the probing problem as a Markov decision process (MDP) defined by states, actions, transition dynamics, and rewards.

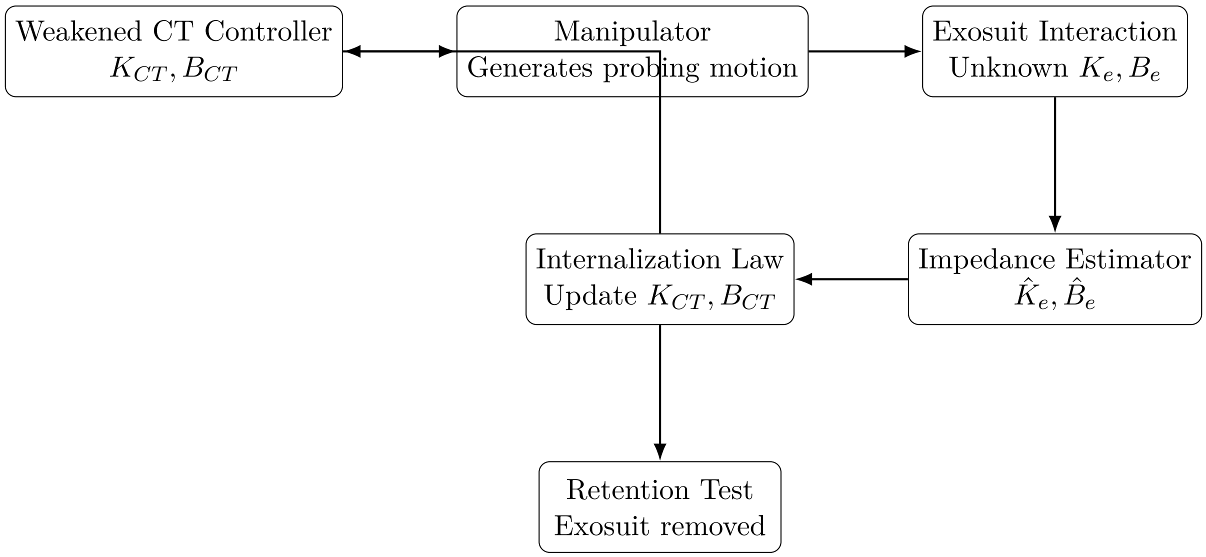


Figure 2: Proposed rehabilitation-inspired robotic learning framework. A weakened CT controller drives the manipulator, which probes the exosuit. Estimated exosuit stiffness and damping are internalized into the CT controller, and retention is evaluated after exosuit removal.

#### 17.3.1 State Space

The RL state should encode the current mechanical and estimation context. A suitable state vector is

$$s_t = \left[q_t,\ \dot{q}_t,\ \hat{K}_e(t),\ \hat{B}_e(t),\ \mathrm{vec}(P_t),\ e_t,\ \tau_t\right], \tag{63}$$

where:

- $q_t$ is the joint angle,
- $\dot{q}_t$ is the joint velocity,
- $\hat{K}_e(t)$ and $\hat{B}_e(t)$ are the current impedance estimates,
- $P_t$ is the estimator covariance or uncertainty matrix,
- $e_t$ is the task tracking error,
- $\tau_t$ is the interaction torque.

If a lower-dimensional state is preferred, one may use

$$s_t = [q_t, \dot{q}_t, \hat{K}_e(t), \hat{B}_e(t), u_t], \tag{64}$$

where $u_t$ is a scalar uncertainty measure, such as $u_t = \mathrm{tr}(P_t)$.

#### 17.3.2 Action Space

The RL action determines the probing command. Several formulations are possible.

**Amplitude-frequency probing.** The agent selects the amplitude and frequency of a probing motion:

$$a_t = [A_t, \omega_t]. \tag{65}$$

A corresponding probing reference can be

$$q_{\text{probe}}(t) = q_{\text{ref}}(t) + A_t \sin(\omega_t t). \tag{66}$$

**Torque-based probing.** Alternatively, the agent directly selects an excitation torque:

$$a_t = \tau_{\text{probe}}(t). \tag{67}$$

**Motion primitive selection.** A more structured approach is to let the agent choose among predefined probing primitives, such as:

- slow flexion-extension,
- sinusoidal oscillation,
- velocity pulse,
- small perturbation around equilibrium.

In that case, $a_t$ indexes a discrete set of probing behaviors.

#### 17.3.3 Transition Dynamics

The system evolves according to the manipulator dynamics, exosuit interaction, estimator updates, and CT controller response. RL does not require an explicit analytical transition model if a model-free method is used, but conceptually the dynamics are

$$s_{t+1} \sim p(s_{t+1} \mid s_t, a_t). \tag{68}$$

The transition includes:

1. application of probing action $a_t$,
2. physical interaction with the exosuit,
3. collection of force-motion measurements,
4. estimator update producing $\hat{K}_e(t+1)$, $\hat{B}_e(t+1)$, and $P_{t+1}$.

### 17.4. Reward Design

The probing policy should maximize information gain while minimizing unsafe or inefficient interaction. A suitable reward is

$$r_t = \alpha \, \text{InfoGain}_t - \beta \, \|\tau_t\|^2 - \gamma \, \text{SafetyCost}_t - \delta \, \|e_t\|^2, \tag{69}$$

where:

- $\text{InfoGain}_t$ rewards reduction in parameter uncertainty,

- $\|\tau_t\|^2$ penalizes excessive probing effort,
- $\mathrm{SafetyCost}_t$ penalizes unsafe motions or torque limits,
- $\|e_t\|^2$ penalizes excessive deviation from the desired task.

A practical information-gain term is

$$\mathrm{InfoGain}_t = \mathrm{tr}(P_t) - \mathrm{tr}(P_{t+1}), \tag{70}$$

so that reward is positive when estimator uncertainty decreases.

Thus, a more explicit reward becomes

$$r_t = \alpha\big(\mathrm{tr}(P_t) - \mathrm{tr}(P_{t+1})\big) - \beta\|\tau_t\|^2 - \gamma\,\mathrm{SafetyCost}_t - \delta\|e_t\|^2. \tag{71}$$

### 17.5. Safety-Aware Reward Terms

To ensure rehabilitation-consistent probing, safety penalties can be introduced when joint limits, velocity bounds, or torque bounds are violated. For example,

$$\mathrm{SafetyCost}_t = c_q\,\mathbf{1}_{q_t\notin\mathcal{Q}} + c_{\dot q}\,\mathbf{1}_{\dot q_t\notin\mathcal{V}} + c_\tau\,\mathbf{1}_{\tau_t\notin\mathcal{T}}, \tag{72}$$

where:

- $\mathcal{Q}$ is the safe joint-angle range,
- $\mathcal{V}$ is the safe velocity range,
- $\mathcal{T}$ is the safe torque range.

A smooth alternative is to use quadratic barrier penalties:

$$\mathrm{SafetyCost}_t = c_q\,\mathrm{dist}(q_t,\mathcal{Q})^2 + c_{\dot q}\,\mathrm{dist}(\dot q_t,\mathcal{V})^2 + c_\tau\,\mathrm{dist}(\tau_t,\mathcal{T})^2. \tag{73}$$

### 17.6. Objective of the RL Agent

The RL agent learns a probing policy

$$\pi_\theta(a_t \mid s_t), \tag{74}$$

parameterized by $\theta$, that maximizes the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T}\gamma_{\mathrm{RL}}^t r_t\right], \tag{75}$$

where $\gamma_{\mathrm{RL}} \in (0,1)$ is the RL discount factor.

The learned policy should therefore generate probing motions that:

- excite the exosuit sufficiently to reveal $K_e$ and $B_e$,
- reduce estimator uncertainty quickly,
- avoid unsafe or overly large interactions,
- remain compatible with the CT-controlled movement objective.

### 17.7. Connection to Parameter Estimation

The RL policy improves the quality of collected data rather than directly estimating parameters itself. The parameter estimator still performs the identification step:

$$\hat{\theta}_{t+1} = \mathcal{E}(\hat{\theta}_t, y_t, \phi_t), \tag{76}$$

where

$$\hat{\theta}_t = \begin{bmatrix} \hat{K}_e(t) \\ \hat{B}_e(t) \end{bmatrix}. \tag{77}$$

For instance, under recursive least squares,

$$\hat{\theta}_{t+1} = \hat{\theta}_t + L_t(y_t - \phi_t^\top \hat{\theta}_t), \tag{78}$$

where $L_t$ is the estimator gain.

Hence, RL and estimation play complementary roles:

- RL chooses how to probe,
- the estimator computes what the exosuit parameters are from the resulting measurements.

### 17.8. Recommended RL Algorithms

Because the probing command is naturally continuous, suitable RL methods include:

- Deep Deterministic Policy Gradient (DDPG),
- Twin Delayed DDPG (TD3),
- Soft Actor-Critic (SAC),
- Proximal Policy Optimization (PPO), if the probing command is parameterized suitably.

If the probing strategy is chosen from a finite set of motion primitives, discrete-action methods such as Deep Q-Networks (DQN) may also be used.

Among these, SAC is especially attractive because it balances reward maximization and exploration, which is valuable when informative probing requires persistent but safe exploration.

### 17.9. Integration into the Full Framework

The full loop with RL-based probing is:

1. initialize weakened CT parameters $K_{CT}, B_{CT}$,
2. observe current state $s_t$,
3. RL policy selects probing action $a_t$,
4. manipulator executes the probing motion,
5. exosuit interaction produces torque-motion data,
6. estimator updates $\hat{K}_e, \hat{B}_e$ and uncertainty $P_t$,
7. reward is computed from information gain, effort, safety, and task error,
8. RL policy is updated,
9. estimated exosuit parameters are used to update CT parameters.

This closes the loop between exploration, identification, and internalization.

### 17.10. Compact Summary Equation

The overall probing-learning loop may be summarized as

$$a_t \sim \pi_\theta(\cdot \mid s_t), \tag{79}$$

$$(q_t, \dot{q}_t, \tau_t) \longrightarrow (\hat{K}_e(t), \hat{B}_e(t), P_t), \tag{80}$$

$$r_t = \alpha\big(\mathrm{tr}(P_t) - \mathrm{tr}(P_{t+1})\big) - \beta\|\tau_t\|^2 - \gamma\,\mathrm{SafetyCost}_t - \delta\|e_t\|^2, \tag{81}$$

$$K_{CT}(t+1) = K_{CT}(t) + \eta_K w_t \hat{K}_e(t), \tag{82}$$

$$B_{CT}(t+1) = B_{CT}(t) + \eta_B w_t \hat{B}_e(t). \tag{83}$$

This shows clearly that RL optimizes *how to probe*, while the controller update determines *how learned assistance is internalized.*

## 18. Probable Approaches Given the Current Implementation

The system is already far enough along in the repository that the next step does not need to be "build the model from scratch." A more useful framing is to choose which scientific claim to pursue first. The existing implementation supports several plausible paths.

### 18.1. Approach 1: Direct Impedance-Shaping Evidence

This is the most immediate path. Use the implemented PyDrake exosuit model in the main tendon-with-exo PyDrake script

`script_cup_manipulator_pendulam_tendon_with_exo_pydrake.py`

and the existing VS Code tasks for exo ON/OFF, disturbance-synchronized, and reactive activation.

The core experiment is:

$$\tau_{\mathrm{exo}} \approx -K_{\mathrm{exo,eff}}(q_2 - q_{2,a}) - D_{\mathrm{exo,eff}}\dot{q}_2, \tag{84}$$

$$K_{\mathrm{exo,eff}} = 2k_{\mathrm{exo}}r_{\mathrm{exo}}^2, \qquad D_{\mathrm{exo,eff}} = 2b_{\mathrm{exo}}r_{\mathrm{exo}}^2. \tag{85}$$

Compare tracking and disturbance rejection under identical axes for:

- exo off / transparent baseline,
- exo on during the disturbance window,
- reactive exo activation triggered by tracking error,
- multiple trajectories: circle, rectangle, figure-eight, and line.

The likely first claim is modest but defensible: the exosuit behaves like a programmable impedance layer that improves disturbance rejection and produces a clearer state-force relationship. This path is best for a first result because it is already implemented, easy to visualize, and does not require proving a global learning theorem.

### 18.2. Approach 2: Parameter-Identification View of Learning

The second path treats the exosuit as something the controller/human can probe. From logged trajectories, estimate effective stiffness and damping by fitting

$$\tau_{\mathrm{int}}(t) \approx -\hat{K}\,(q_2(t) - q_{2,a}) - \hat{D}\,\dot{q}_2(t). \tag{86}$$

Then use the estimates to update a weak controller:

$$K_{P,t+1} = K_{P,t} + \eta_K(\hat{K} - K_{P,t}), \tag{87}$$

$$K_{D,t+1} = K_{D,t} + \eta_D(\hat{D} - K_{D,t}). \tag{88}$$

This is not meant to claim that the brain literally estimates numerical gains. Rather, it provides an engineering analogue for the rehabilitation idea: repeated interaction with a structured impedance can create a more learnable feedback signal. This approach is especially useful if the paper is framed around motor learning or rehabilitation rather than pure controller synthesis.

### 18.3. Approach 3: Local Contraction / CCM Certification

The contraction-theory implementation already includes MATLAB scripts for comparing no-exo and with-exo cases:

- `script_14e_cupex_CCM_block_LMI_no_exo.m`,
- `script_14f_cupex_CCM_block_LMI_with_exo.m`,
- `script_14g_cupex_CCM_original.m`.

This path asks whether the passive exosuit terms make the linearized or sampled nonlinear dynamics easier to certify as contracting. Conceptually, the exo adds negative stiffness/damping-like terms to the elbow dynamics, so the certified contraction rate may improve:

$$\lambda_{\max}^{\text{exo}} > \lambda_{\max}^{\text{no exo}}. \tag{89}$$

This is the cleanest theory-facing approach. The limitation is that the result is local or sampled over a chosen region, not a universal statement about all human-exosuit behavior. That limitation is acceptable if stated clearly: the exosuit increases the size or strength of a certifiable stable region around the task dynamics.

### 18.4. Approach 4: C3M Residual Controller on Top of Computed Torque

The C3M path is already represented by the PyTorch systems and MATLAB deployment scripts:

- `C3M/systems/system_CUPMANIP_SEA_EXO_TRACK.py`,
- `C3M/systems/system_CUPMANIP_SEA_EXO_LINEXO.py`,
- `script_14h_cupex_C3M_SEA_exo.m`,
- `script_14i_cupex_C3M_exo_embedded.m`.

The practical insight from today is that the raw input matrix can be state-dependent through the manipulator inertia. The constant-$B$ C3M variants freeze $B$ at the operating point so that the difficult C2 compatibility term is removed from the training burden:

$$B(x) \to B(x_*), \qquad \frac{\partial B}{\partial x} = 0. \tag{90}$$

The likely role of C3M in the near term is not to replace computed torque completely. A safer use is as a residual stabilizing policy around a known baseline:

$$u(t) = u_{\text{CT}}(t) + u_{\text{C3M}}(x(t), x_*(t)). \tag{91}$$

This approach can answer whether learned contraction-style feedback improves robustness beyond CT alone. It is higher risk than Approach 1 or 3 because training/deployment mismatch, motor-state chatter, and residual scaling can dominate the result.

### 18.5. Approach 4b: AI / RL / Learning-Based Implementation Options

AI and learning methods should be used as structured add-ons to the model-based controller, not as an immediate replacement for the physics. The repository already contains the right starting points: the Isaac Sim PPO residual environment in `rl/envs/manipulator_residual_env.py`, the training script `rl/train_ppo_residual.py`, the evaluator `rl/eval_ppo_residual.py`, and the MATLAB scaffold `script_14d_cupex_RL_residual.m`.

The safest learning architecture is residual learning:

$$\tau_{\text{cmd}}(t) = \tau_{\text{CT}}(t) + \Delta\tau_\theta(o_t), \qquad \|\Delta\tau_\theta\|_\infty \le \tau_{\text{res,max}}. \tag{92}$$

Here CT remains the stabilizing backbone, while the learned policy handles effects that are hard to model exactly: SEA phase lag, cable slack/compliance, friction, unmodelled exo coupling, and trajectory-dependent tracking error.

The implemented RL observation already has a useful structure:

$$o_t = [q_1, q_2, \dot{q}_1, \dot{q}_2, e_x, e_y, \delta, \dot{\delta}, F_c, \tau_{2,\text{CT}}, \tau_{2,\text{SEA}}, \tau_m, e_{\tau_1}, e_{\tau_2}]^\top. \tag{93}$$

This observation is appropriate because it includes not only kinematic tracking error, but also the actuator-side quantities that reveal why CT may fail: spring extension, cable force, applied torque, motor torque, and torque-tracking error.

Several learning routes are plausible:

1. **Residual PPO policy.** Train $\Delta\tau_\theta$ with PPO in Isaac Sim. Reward should penalize EE error, residual effort, acceleration/smoothness, and torque-tracking mismatch:

   $$r_t = -w_e\|e_{\text{EE}}\|^2 - w_u\|\Delta\tau_\theta\|^2 - w_a\|\ddot{q}\|^2 - w_\tau\|\tau_{\text{des}} - \tau_{\text{applied}}\|^2. \tag{94}$$

   This is the most direct use of the existing `rl/` code.

2. **Gain adaptation policy.** Instead of outputting torque directly, learn a slower policy that adjusts impedance or controller gains:

   $$[K_P, K_D, K_{\text{exo}}, D_{\text{exo}}]_{t+1} = [K_P, K_D, K_{\text{exo}}, D_{\text{exo}}]_t + \Delta_\theta(o_t). \tag{95}$$

   This is more interpretable for the rehabilitation story because the learned action changes "how stiff" and "how damped" the interaction feels rather than injecting arbitrary torque.

3. **Learned exo activation policy.** Replace hand-coded timed/reactive activation with a classifier or policy:

   $$a_{\text{exo}}(t) \in \{0,1\}, \qquad \Delta\theta_{\text{exo}}(t) = \pi_\theta(o_t). \tag{96}$$

   This would learn when assistance should turn on, how much co-contraction offset to apply, and when to return to transparent mode.

4. **Supervised imitation from an oracle.** Generate data from CT, CCM, or hand-tuned exo-on/off experiments and train a smaller policy to imitate the best action. This can be lower variance than RL and gives a clear baseline before PPO.

5. **System-identification network.** Learn $\hat{K}, \hat{D}$, friction, or SEA lag parameters from trajectory windows, then feed those estimates into a model-based controller. This keeps the learned model interpretable and directly tied to the meeting hypothesis.

6. **Offline model-based learning.** Use simulation logs to fit a disturbance or residual dynamics model
$$x_{t+1} \approx f_{\text{model}}(x_t, u_t) + d_\theta(x_t, u_t), \tag{97}$$
then use the learned residual model for MPC, gain scheduling, or safer policy search.

The recommended implementation order is conservative:

1. run CT-only and CT+exo baselines first,
2. train residual PPO with small torque bounds and compare against CT-only,
3. add exo-aware observations and reward terms if the policy ignores cable/exo lag,
4. move from torque residuals to gain/impedance adaptation for interpretability,
5. finally test learned exo activation and catch-trial after-effects.

The paper framing should emphasize that AI is a tool for discovering or tuning the feedback-shaping rule. The scientific claim remains physical: the exosuit creates a structured impedance channel; learning methods determine how to use that channel effectively.

### 18.6. Approach 5: Full Coupled Exosuit-State Certification

The most physically faithful C3M path is the full augmented-state model, represented by

`C3M/systems/system_CUPMANIP_SEA_EXO_FULL.py` and `script_14j_cupex_C3M_exo_full.m`.

Here the exo motor states are included in the learning/certification state, e.g.

$$x_{10} = [q_1, q_2, \dot{q}_1, \dot{q}_2, \theta_m, \dot{\theta}_m, \theta_{mR}, \dot{\theta}_{mR}, \theta_{mL}, \dot{\theta}_{mL}]^\top. \tag{98}$$

This avoids pretending that the exosuit is only an algebraic spring-damper torque. The tradeoff is complexity: the network is larger, the training problem is harder, and the explanation is less transparent. This should probably be treated as a later robustness check, not the first paper's main result.

### 18.7. Approach 6: Human / Rehabilitation Translation

Once the simulation story is stable, the human-facing interpretation can be tested with catch-trial logic. The simulation analogue is:

1. Train or adapt the weak controller with exo impedance active.
2. Remove or reduce the exo impedance suddenly.
3. Check whether the controller shows an after-effect or retained compensation.

In a human study, the same logic would ask whether participants adapt their motion after repeated exposure to a structured exosuit impedance. This is the strongest rehabilitation framing, but it should come after the simulation has a clean effect size and a clear mechanistic story.

### 18.8. Recommended Order

The lowest-risk path is:

1. establish exo ON/OFF disturbance-rejection plots with identical axes,
2. quantify RMS error, peak error, settling time, and effective damping,
3. run the CCM no-exo vs with-exo comparison to support the stability interpretation,
4. add the parameter-identification story as the bridge to learning,
5. use C3M residual control only after the baseline and CCM results are clean,
6. reserve full 10D exo-state C3M and human catch trials as later-stage validation.

Therefore, the probable first paper should not overclaim that the exosuit directly teaches the brain an exact $K_P, K_D$ controller. A stronger and more defensible claim is: a programmable exosuit can shape interaction impedance, improve local stability/disturbance rejection, and provide an interpretable physical channel that can be used for controller adaptation or rehabilitation-oriented learning.

## 19. Open Conceptual Challenge

One of the biggest unresolved questions was:

> How exactly do we map the exosuit-provided physical feedback into a useful internal adaptation of the controller/brain?

Several versions of the challenge were discussed:

- exosuit impedance is not the same thing as neural control gains,
- parameter estimates do not directly explain perception,
- a strong engineering mapping may not align with human interpretation,
- and there may need to be an intermediate representation or "language" between the two domains.

This was variously described as:

- a mapping problem,
- an interface problem,
- a representation problem,
- or even a "language" problem.

One tentative practical compromise discussed was:

- initially assume a simple direct mapping from learned exosuit parameters to controller parameter changes,
- use that for simulation,
- and improve the interpretation layer later.

## 20. Potential Role of AI / Learning Methods

There was brief discussion of whether modern AI or learning-based methods might help.

Possible directions mentioned included:

- reinforcement learning,
- data-driven adaptation,
- human-in-the-loop learning,
- and behavioral modeling.

However, this appeared exploratory rather than a final decision. There was also caution about:

- complexity,
- interpretability,
- and whether a simpler model-based route should be tried first.

## 21. Short-Term Technical Action Items

The following near-term tasks were either explicitly stated or strongly implied:

1. **Simplify the disturbance input:** replace sinusoidal disturbance with a step or pulse for clearer interpretation.
2. **Overlay plots:** compare exosuit on/off responses on the same axes.
3. **Check scaling:** ensure differences are not due to inconsistent plot scales.
4. **Add omitted model components:** include any missing passive/mechanical terms if needed.
5. **Build a toy coupled-system model:** represent the arm/controller and exosuit as interacting second-order systems.
6. **Probe parameter regression:** test whether exosuit stiffness/damping can be estimated from interaction data.
7. **Map identified parameters to weak controller gains:** try a simple heuristic update rule first.
8. **Test critical-damping target behavior:** see whether the system can learn or be driven toward critically damped response.
9. **Clarify contraction-control role:** decide whether contraction is central theory, supporting intuition, or only a local design tool.

## 22. Possible Paper Directions

There was some discussion about publication strategy and what the first paper should actually claim.

Possible paper angles include:

### 22.1. Paper Angle A: Formulation / Theory

Focus on:

- the hypothesis that exosuit-shaped impedance provides a learning-friendly feedback channel,
- mathematical formulation,
- simplified analysis,
- and simulation evidence.

### 22.2. Paper Angle B: Design / Platform

Focus on:

- exosuit mechanism,
- cable-based stiffness modulation,
- and use as a programmable rehabilitation interface.

### 22.3. Paper Angle C: Simulation + Human Pilot

Focus on:

- simulation formulation,
- parameter learning idea,
- and preliminary human testing or catch trials.

There was also discussion of aiming for a submission before the end of the year / before Christmas, with some uncertainty about venue and risk.

## 23. Notes on Unclear Terms from the Transcript

A few transcript fragments were likely corrupted or ambiguous. The following are **best-effort interpretations**:

- "KPKD" was interpreted as proportional/derivative gain discussion.
- "CCM" was interpreted as control contraction metrics.
- "residual controller" appears to refer to an additional controller layered over an existing baseline controller.
- "computer talk" likely meant *computed torque*.
- "contraction region" likely referred to learning or operating in a stable/convergent regime.
- "back drivable mode" appears to refer to exosuit transparency or low-impedance interaction mode.

## 24. Non-Technical / Lab Logistics

The meeting also included unrelated but important practical discussion:

### 24.1. Workspace and Lab Setup

- There was interest in making the space feel like a proper lab.
- Whiteboards were suggested as useful additions.
- A possible count/arrangement of desks and available space was discussed.
- It was noted that more visible whiteboard space might help technical discussion and engagement.

### 24.2. Meeting Room

- A nearby meeting room was mentioned.
- It appears to be bookable.
- Its location was described relative to the office entrance.

### 24.3. Miscellaneous Side Conversation

There were also short unrelated fragments about:

- printing/layout issues,
- fish/shrimp setup,
- kitchen/lab usage,
- and general office organization.

  These do not appear central to the technical project.

## 25. Concise Summary

In compact form, the meeting's technical direction can be summarized as follows:

1. Model the human/controller as a weak dynamical system.
2. Model the exosuit as providing programmable impedance (stiffness/damping).
3. Let the controller interact with the exosuit and observe force/state relationships.
4. Estimate or infer the exosuit dynamics from this interaction.
5. Use that interaction as a structured feedback signal to guide controller adaptation.
6. Test whether this creates faster or more stable learning than unaided interaction.
7. Connect the result to rehabilitation: the exosuit may both assist movement and improve relearning.

## 26. Immediate Next-Step Version

A minimal actionable version of the plan is:

1. Use a simple second-order toy model.
2. Add exosuit stiffness and damping.
3. Apply a step disturbance.
4. Compare responses with exosuit on/off.
5. Attempt parameter identification from interaction data.
6. Use identified values to heuristically update weak controller gains.
7. Check whether the result moves the system toward a more critically damped response.

## End Note

These notes reflect a cleaned, structured reconstruction of a noisy conversation. They are suitable as a working record, but any equations, terminology, and paper claims should be refined against the actual model and code implementation.